

Surrogate-Conditioned Benchmark Fragility in Block-Scale Urban Energy Design: A Comparative Study of Policy Learning and NSGA-II

Response to the Reviewers

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Dear Editors,

We sincerely thank the Editors and Reviewers for the careful evaluation of our manuscript and for the constructive suggestions. The comments have been very helpful in improving both the technical positioning and the presentation of the paper. In the revised submission, we re-examined the benchmark framing, clarified the methodological limits of the current evidence, updated the manuscript and appendix accordingly, and improved the organization and presentation of the figures, tables, and supporting explanations.

Below we provide a point-by-point response to all reviewer comments. For ease of inspection, the revised manuscript highlights the main textual changes and marks the corresponding locations in the margin notes.

We hope that the revised manuscript and the detailed responses below adequately address the concerns raised during review.

With kind regards,

Zhi Wu, on behalf of the authors

Manuscript information

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Reviewer I

Comment I.1:

The manuscript presents an ambitious framework that integrates surrogate modeling and deep reinforcement learning for urban morphology and energy optimization. However, several critical issues currently limit the robustness and interpretability of the reported findings. Substantial changes are required before the manuscript can be considered for publication.

Answer: We sincerely thank the Reviewer for the careful reading and for the constructive and technically demanding comments. We agree that the earlier manuscript framed the optimizer comparison too optimistically and did not communicate the methodological limits of the current evidence with sufficient clarity. We therefore used these comments to rethink not only individual paragraphs, but the overall presentation of the paper. The revised paper is no longer presented as a DRL-superiority study. Instead, it is framed as a surrogate-conditioned benchmark-fragility study in which the selected 2000-sample surrogate, the checkpoint-sensitivity audit, the same-budget CMA-ES follow-up, the independent candidate reevaluation, and the bounded physical probe are used together to support a more cautious interpretation. We respond point by point below.

In practical terms, this revision is not a light editorial cleanup. We revised the title, abstract, contribution list, benchmark section, discussion, and conclusion; rebuilt the main benchmark figure set; expanded Appendix A with scale-study and surrogate-fidelity diagnostics; expanded Appendix B with checkpoint-sensitivity, independent reevaluation, and bounded physical-probe evidence; and systematically corrected notation, figure labels, captions, units, and abbreviation handling. Our intent in the present response letter is therefore to make these changes visible rather than simply state that the manuscript has been revised.

Comment I.2:

1. The study relies exclusively on simulated outputs to generate the dataset for surrogate modeling and DRL optimization, without any empirical or independent external validation to assess the realism of the simulation results.

Answer: We agree that full empirical or full-stack external validation is still unavailable, and we now state that limitation explicitly. At the same time, the revision strengthens the validation side in two bounded ways.

First, Appendix B now reports an independent deterministic reevaluation of the selected DDPG and NSGA-II candidates, so the final comparison is no longer interpreted solely through optimization-time surrogate predictions.

Second, we added a small-batch physical probe on representative candidates. Under the current best-aligned project workflow, EUIt shows the strongest agreement, EG remains a simplified but directionally stable proxy, and H is now checked through a January 20 windowsill-style direct-sun-hours workflow that is much closer to the manuscript definition than the earlier proxy. We also state explicitly that this does not constitute full physical closure.

Accordingly, the point is addressed through stronger bounded validation and much clearer framing, but not by claiming that external validation is fully closed. We appreciate this criticism because it helped us avoid presenting the revised package as more complete than it really is.

Revised manuscript excerpt

Revised evidence-scope text:

“The benchmark comparison is conducted on the selected 2000-sample surrogate derived from the analytic response-generation workflow, whereas direct physical probing is currently available only for a small set of representative candidates. The results below therefore combine a full surrogate-space benchmark with a bounded physical cross-check rather than a fully symmetric end-to-end physical rerun.”

Revised abstract text:

“A small-batch physical probe adds bounded external support: EUIt aligns most consistently, the current EG check remains weaker but directionally stable, and a revised windowsill-style direct-sun-hours check places H in the same broad range as the surrogate values, albeit with weaker stability than EUIt.”

Representative candidate	EUIt	EG	H
DDPG / balanced (surrogate)	66.000	2.850	7.850
DDPG / balanced (physical probe)	74.379	2.637	7.525
NSGA-II (surrogate)	69.443	2.613	7.569
NSGA-II (physical probe)	74.057	2.828	6.075

Representative values summarized from the bounded physical probe now reported in Appendix B.



Revised four-method physical-probe figure included in Appendix B to make the new validation-mode evidence directly visible.

Comment I.3:

2. *The claimed superiority of the DRL framework over NSGA-II may be methodologically biased, as the utopian-point-based reward explicitly guides the search toward a predefined ideal region, whereas NSGA-II follows a standard Pareto-based formulation emphasizing diversity.*

Answer: We agree, and this concern is now treated as part of the paper’s central interpretation. In the revised manuscript, the reward is described explicitly as a scalarized weighted-distance objective around a utopian point, not as a direct Pareto-learning formulation. We also removed the earlier superiority claim. On the selected highest-accuracy 2000-sample surrogate, fair-budget NSGA-II remains stronger than all three DDPG scenarios in objective summaries, Hypervolume, IGD, and preference-aligned utility. The DDPG scenarios are retained as evidence of preference-conditioned search bias on a surrogate checkpoint, not as proof of Pareto superiority.

To make this shift visible rather than merely verbal, the revised manuscript now reports the benchmark ordering directly through the main benchmark table and figure set. In particular, Table 7 and Fig. 9 show that NSGA-II remains strongest on the

selected surrogate, while the DDPG scenarios remain weaker and partially overlapping. We also added the main-text budget-accounting table so that the comparison protocol is explicit rather than implicit. This comment helped us replace a defensively framed response with a clearer statement of what the benchmark actually shows.

Revised manuscript excerpt

Revised contribution text:

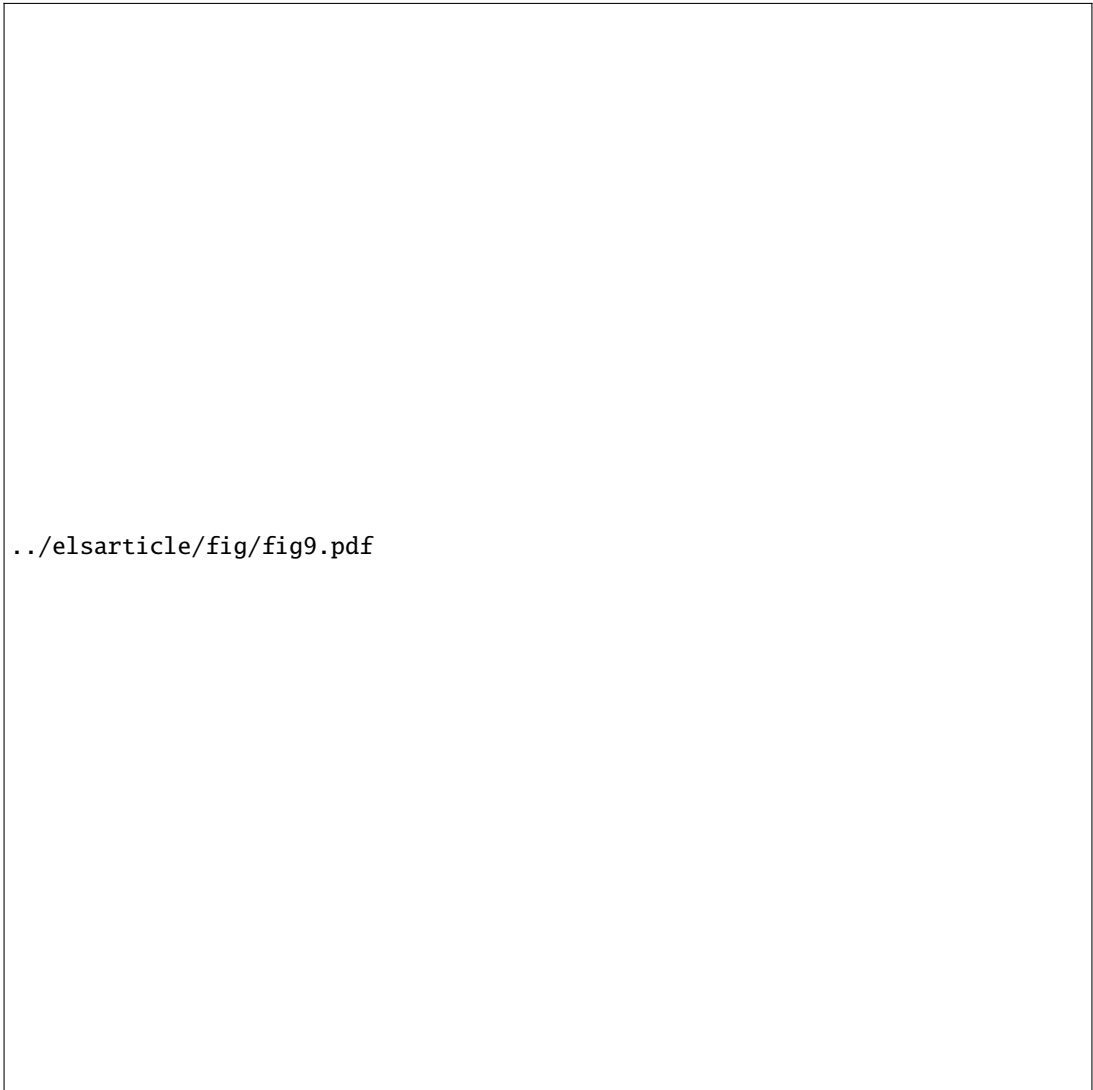
“This study shows that, on the selected highest-accuracy surrogate, fair-budget NSGA-II remains stronger than all three DDPG scenarios in archive-level Pareto metrics and preference-aligned utility, while the DDPG runs retain substantial late-stage instability.”

Revised benchmark text:

“Rather than claiming superiority of DRL over NSGA-II, the available evidence indicates that DDPG does not match fair-budget NSGA-II on the selected surrogate.”

Method	HV	IGD
NSGA-II	1.3310	0.1068
Balanced DDPG	0.3474	0.7196
Saving DDPG	0.6252	0.5491
Generation DDPG	1.0107	0.2205

Main benchmark numbers now reported in the revised manuscript for the selected 2000-sample surrogate.



Revised benchmark summary figure showing scenario-wise objective summaries and preference-aligned utility under the selected 2000-sample surrogate.

Comment I.4:

3. The surrogate model is trained on only 500 samples in a 12-dimensional continuous design space, which may be sufficient for regression but is likely insufficient to robustly support surrogate-guided reinforcement learning optimization, particularly in extreme or sparsely sampled regions.

Answer: We agree that the original presentation did not make the dataset scale and its implications sufficiently clear. The revised manuscript now distinguishes between the initial 500-sample baseline and the expanded 500/1000/1500/2000-sample scale study. The final optimizer benchmark is conducted on the selected 2000-sample surrogate, and Appendix A now reports the scale-study diagnostics, cross-validation summary, and extreme-region error diagnostics together. We also retained an explicit limitation stating that even the selected 2000-sample surrogate still approximates a 12-dimensional continuous design space and should not be interpreted as closing extrapolation risk in sparse or extreme regions.

This change is also reflected in the current figure set. The revised manuscript no longer asks the reader to accept the final surrogate by default. Instead, the scale-study figure makes clear why the selected 2000-sample surrogate was retained for the final comparison and how the final benchmark is tied to an explicit model-selection procedure. This was an important improvement because the earlier draft left too much of that selection logic implicit.

Revised manuscript excerpt**Revised methodology text:**

“A dataset-scale study was carried out over 500, 1000, 1500, and 2000 morphologies, and the final optimizer comparison uses the most accurate surrogate identified from that study.”

Revised limitation text:

“Even with 2000 simulated cases, the surrogate still approximates a 12-dimensional continuous design space and therefore remains vulnerable in sparse or extreme regions.”

../elsarticle/fig/fig11.pdf

Scale-study diagnostic figure now included in the manuscript to make the final surrogate selection traceable across the 500/1000/1500/2000-sample regimes.

Comment I.5:

4. The literature review remains largely descriptive and lacks critical synthesis from an energy systems perspective, framing the research gap primarily as an algorithmic improvement rather than addressing unresolved challenges in energy system design and decarbonization.

Answer: We agree and revised the Introduction accordingly. The current version now links the methodological contribution more directly to block-scale urban-energy design under decarbonization constraints and to the practical role of surrogate-assisted optimization as a screening layer before full physical validation. This allows the research gap to be stated as a domain-relevant methodological question: when optimization is surrogate-conditioned, benchmark reliability itself becomes part of the urban-energy design problem.

Revised manuscript excerpt

Revised Introduction text:

“Large comparative sweeps over urban-energy design strategies are often bottlenecked by expensive simulation workflows, so surrogate-assisted optimization is increasingly used as a screening layer before full physical validation. However, once the benchmark itself is surrogate-conditioned, optimizer reliability becomes part of the research problem: if a learning-based method is unstable, checkpoint-sensitive, or difficult to compare fairly, its practical value for urban energy design remains uncertain.”

Comment I.6:

5. On Page 17 (Line 55), the manuscript claims that the objective space contains complex, nonlinear valleys of high performance. However, the analysis presented on Pages 16-17 considers only linear relationships (Pearson Correlation). The absence of nonlinear interaction analysis is not adequately explained.

Answer: We agree that the earlier wording was too strong. We removed the unsupported “nonlinear valleys” claim from the main text. In the revised manuscript, the correlation analysis is used only as a first-order interpretive tool, while nonlinear response information is reported in Appendix B as supporting surrogate-based diagnostics. This narrows the claim to what is actually shown by the available evidence.

Revised manuscript excerpt

Revised appendix text:

“To complement the linear correlation results in the main text, the appendix now reports representative surrogate-based nonlinear response profiles for OSR→EUIt, FAR→EG, SVF→H, and $\theta \rightarrow H$.”

The stronger “nonlinear valleys” wording has been removed from the main manuscript.

Comment I.7:

6. Given the 12-dimensional continuous action space, the training duration (200 episodes $\times$ 40 steps) appears relatively short. Stronger evidence or justification is required to demonstrate that the DRL agents have adequately converged.

Answer: We agree. The final manuscript no longer relies on the earlier short-horizon presentation. Instead, the current training-dynamics evidence comes from guarded 600-episode multi-seed reruns with 20 seeds per scenario. More importantly, we no longer claim stable asymptotic convergence. The revised interpretation is intentionally narrower: policy learning can reach high-performing surrogate regions, but late-stage regression remains frequent, so the role of DDPG in the paper is diagnostic rather than as evidence of robust convergence to a final optimum.

To make this interpretation concrete, the revised text now reports the approximate plateau episodes together with the best-to-final reward gap ratios and substantial-regression fractions. On the selected 2000-sample surrogate, the 20-episode rolling reward reaches 95% of its best observed value around episodes 401, 320, and 398 for the balanced, saving-focused, and generation-focused scenarios, respectively. However, the substantial-regression fractions remain high at 95%, 85%, and 100%, which is why the revised manuscript no longer treats these runs as evidence of clean asymptotic convergence.

Revised manuscript excerpt

Revised training-dynamics text:

“On the selected 2000-sample surrogate, seed-level diagnostics show that the 20-episode rolling reward typically reaches 95% of its best observed value around episodes 401, 320, and 398 for the balanced, saving-focused, and generation-focused scenarios, respectively. At the same time, late-stage regression remains common, with mean best-to-final reward gap ratios around 0.62, 0.71, and 0.71 and substantial-regression fractions of 95%, 85%, and 100% across the three scenarios. These curves therefore support a narrower interpretation: policy learning can reach high-performing surrogate regions, but it does not exhibit stable asymptotic convergence in this setting.”

Comment I.8:

7. The manuscript title does not sufficiently convey the methodological advantage of the proposed approach over conventional NSGA-II-based optimization, despite this comparison being central to the paper’s claimed contribution.

Answer: We addressed this concern by changing the title in a more accurate and more conservative direction. Because the revised evidence does not support an optimizer-superiority claim, the new title foregrounds the methodological issue that the final

paper actually supports, namely benchmark fragility under surrogate conditioning.

Revised manuscript excerpt

Revised title:

“Surrogate-Conditioned Benchmark Fragility in Block-Scale Urban Energy Design: A Comparative Study of Policy Learning and NSGA-II”

Comment I.9:

8. The graphical abstract is of insufficient resolution and does not meet the journal's publication standards.

Answer: We thank the Reviewer for pointing this out. The graphical abstract has been replaced with a higher-resolution publication-quality version aligned with the revised manuscript framing.

In revising this asset, we also made sure that the graphical abstract is consistent with the current final message of the paper. In other words, the replacement was not only a technical export at higher resolution; the content shown in the graphical abstract was also aligned with the revised benchmark-fragility framing so that the visual summary does not continue to imply a stronger optimizer-superiority claim than the revised manuscript supports.

Comment I.10:

9. The definitions of AF and OSR on Page 6 (Lines 45-46) are inconsistent with those in Supplementary Material S-1, which may cause confusion.

Answer: We thank the Reviewer for noting this inconsistency. It has now been corrected. The definitions of AF and OSR are synchronized between the main manuscript and Appendix A, where the former supplementary material has been consolidated.

We agree that such notation drift can undermine reader confidence, especially in a paper that already contains multiple objectives, multiple morphology descriptors, and several layers of appendix material. For that reason, we did not treat this as an isolated wording fix. Instead, we rechecked the morphology-factor definitions across the main text and appendix together and kept only one final definition for each quantity in the revised package.

Revised manuscript excerpt

Revision summary:

The morphology-factor definitions in the main text and Appendix A were checked and aligned so that AF and OSR now use one consistent definition throughout the revised package.

Comment I.11:

10. Several formatting problems are present, including inconsistent or incorrect presentation of " R^2 " on Page 6 and in Supplementary Material S-2.

Answer: We thank the Reviewer for pointing out these formatting inconsistencies. This has now been corrected. The notation for R^2 is standardized in the main manuscript and Appendix A, including the cross-validation and surrogate-fidelity tables.

More broadly, this correction was made as part of a systematic pass over notation consistency rather than as a one-line patch. During this pass, we checked the formatting of statistical symbols, table headers, and metric names together so that the validation-related tables now read as one coherent package.

Revised manuscript excerpt

Examples of revised headings:

“Objective & Mean R^2 & Mean MAE & Std. R^2 & Std. MAE”

“Target & RMSE & MAE & nRMSE & MAPE (%) & R^2 ”

Comment I.12:

11. A table in Supplementary Material S-2 (apparently corresponding to Table V) lacks a table title or caption.

Answer: We thank the Reviewer for this observation. In the revised appendix, all retained tables now have explicit captions, including the surrogate-validation, checkpoint-audit, reevaluation, and physical-probe tables.

We agree that explicit captions are particularly important here because the revised appendix now carries a substantial portion of the technical evidence package. For that reason, we rewrote the captions so that they do not merely name the table, but also indicate what kind of evidence is being reported and how that evidence should be interpreted in the context of the revised manuscript.

Revised manuscript excerpt

Examples of revised captions:

“Additional surrogate-fidelity audit on the cross-validation predictions of the selected 2000-sample surrogate.”

“Checkpoint-sensitivity audit for two 2000-sample surrogate checkpoints.”

“Independent re-evaluation of selected candidates from the highest-accuracy 2000-sample surrogate rerun.”

Comment I.13:

12. Multiple figures and tables lack proper axis labels, units, or legends. In particular, Fig. 7 lacks y-axis explanations and units; Fig. 8, Fig. 11, Fig. 12, and Table VI lack units; Fig. 9 contains partially obscured indicators and missing units; Fig. 13 lacks axis labels and units in some subplots. The relationship between the upper and lower parts of Fig. 2 is unclear and should be clarified in the caption, and Fig. 3 requires clear explanations of colors and line types.

Answer: We have revised the figure and table presentation substantially. Axis labels, units, legends, and caption explanations were checked systematically against the current final figure set. In particular, the benchmark, seed-diagnostics, scale-study, checkpoint-sensitivity, and physical-probe figures now use explicit units and revised captions consistent with the final selected-2000-sample narrative. We also clarified the workflow relation in the caption of Fig. 2 and updated the captions so that readers can interpret each panel without relying on the earlier supplementary package.

This revision required more than cosmetic relabeling. Because the paper’s interpretation changed materially during revision, the figure set also had to be rebuilt so that the visuals matched the final evidence hierarchy. In practice, this meant revising panel labels, legend wording, unit annotations, and caption logic across the benchmark, diagnostics, and validation figures, rather than simply patching one or two isolated omissions.

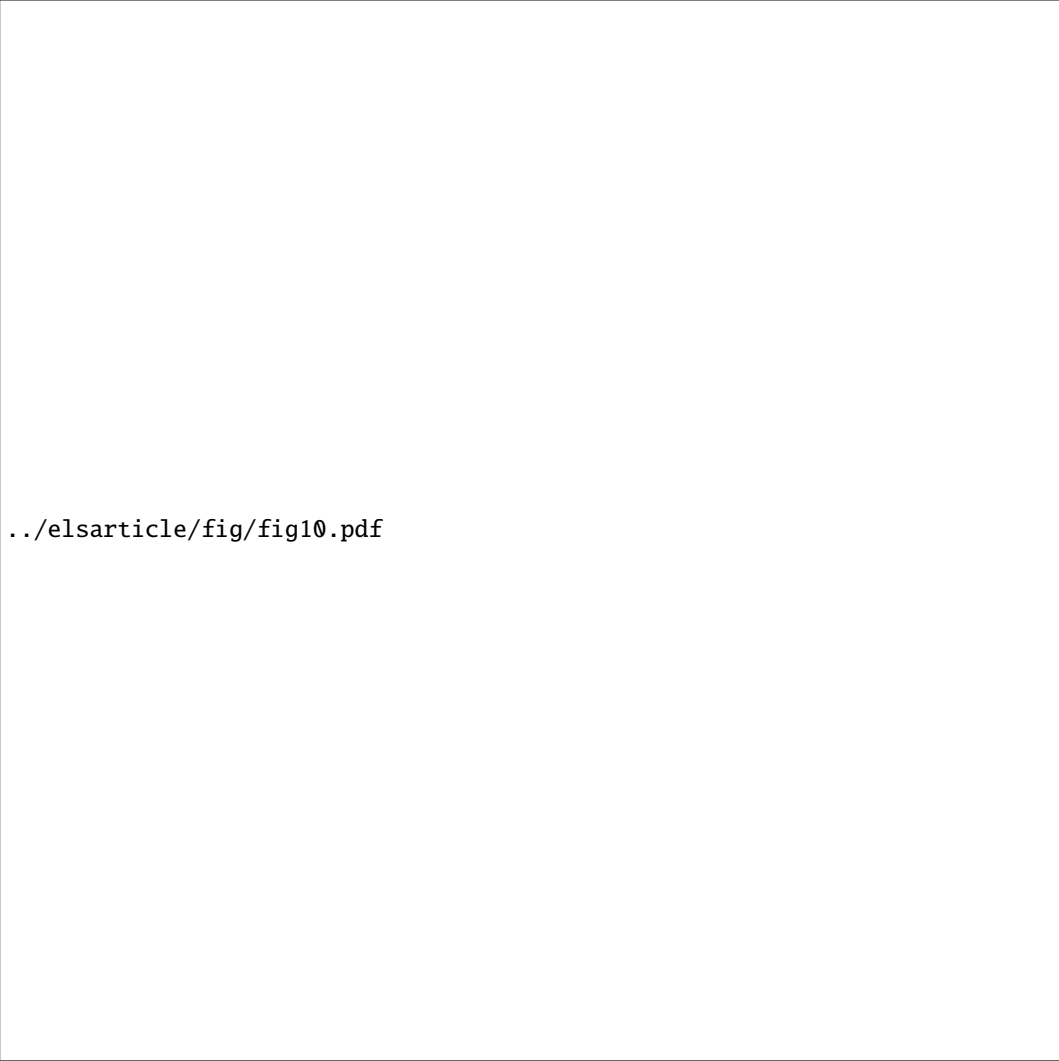
Revised manuscript excerpt

Representative revised captions:

“Comparison across fair-budget NSGA-II and the three DDPG scenarios on the selected 2000-sample surrogate.”

“Seed-level convergence diagnostics on the selected 2000-sample surrogate. Panels report the mean best reward, mean final reward, mean plateau episode, and late-regression fraction across 20 seeds per DDPG scenario.”

“Four-method small-batch comparison between surrogate predictions and the current best-aligned physical probe. Each row shows one representative candidate for DDPG, NSGA-II, CMA-ES, and RandomSearch, with square markers for surrogate values and circular markers for physical-probe values.”



../elsarticle/fig/fig10.pdf

Example of the revised diagnostic figure set, now presented with clearer panel structure, units, and caption support for the selected 2000-sample narrative.

Comment I.14:

13. The manuscript does not include a list of abbreviations, which would improve readability given the extensive use of acronyms.

Answer: We agree with this suggestion. An abbreviations list has now been added near the beginning of the manuscript in order to improve readability.

We found this addition particularly useful after the manuscript was revised into a more diagnostics-heavy version, because the current paper now refers repeatedly to UMFs, EPIs, DDPG, NSGA-II, CMA-ES, HV, IGD, and related terms across the main text and appendix. The abbreviations list is therefore intended not only as a formatting improvement, but also as a practical aid for readability.

Revised manuscript excerpt

Revision summary:

The revised manuscript now includes an “Abbreviations” section listing the main acronyms used in the paper, including UM, UMF, EPI, MOO, DRL, DNN, DDPG, NSGA-II, CMA-ES, SAC, HV, and IGD.

Comment I.15:

14. There are also several formatting issues on Page 6, such as "environments17" and "grid18", which should be corrected.

Answer: We thank the Reviewer for identifying these issues. These spacing and merged-word artifacts have been corrected during the source-level revision of the manuscript. The current version no longer relies on the problematic imported fragments that produced those glitches.

To avoid leaving similar artifacts elsewhere, we did not treat these examples as isolated typos. Instead, we performed a broader source-level cleanup pass on the manuscript so that spacing, merged words, notation rendering, and local text-formatting problems were corrected together.

Reviewer II

Comment II.16:

This paper studies a DRL-DNN framework for block-scale co-optimization of urban morphology and energy performance, using a surrogate model and DDGP agents to search for improved design strategies. The topic is interesting, relevant, and worth investigating. The overall structure of the manuscript is clear, and the figures help communicate the workflow and results. However, several important issues should be addressed before the paper can be considered for publication.

Answer: We sincerely thank the Reviewer for the careful and constructive comments. We especially appreciate the emphasis on methodological positioning, benchmark fairness, surrogate reliability, and the limits of the current validation mode. These comments identified several places where the previous manuscript was not yet sufficiently balanced. In the revised manuscript, the paper no longer claims that DDGP outperforms NSGA-II. Instead, it presents a selected-2000-sample comparison in which fair-budget NSGA-II remains stronger than all three DDGP scenarios, and it interprets that outcome through checkpoint sensitivity, bounded physical probing, and explicit claim limits. We address each point below.

To ensure that the revision is visible at the level of substance rather than only at the level of wording, we also reorganized the manuscript around a clearer evidence structure. This includes the dataset-scale selection logic in Appendix A, the selected-surrogate benchmark figures and budget accounting in the main text, and the checkpoint-sensitivity, reevaluation, and physical-probe diagnostics in Appendix B.

Comment II.17:

The paper is generally well organized, but I recommend using a simpler and more standard numbering system for sections and subsections. This would make the manuscript easier to follow.

Answer: We thank the Reviewer for this helpful suggestion. The revised manuscript now uses a simpler and more standard section and subsection structure. We also consolidated the former supplementary material into the appendix so that the overall organization is easier to follow.

We found this structural cleanup valuable beyond presentation alone. Because the revised manuscript now contains a more layered evidence story, a clearer section hierarchy was necessary to help the reader distinguish among the methodology, the selected-surrogate benchmark, the validation-oriented appendix evidence, and the final limitations discussion. The manuscript now moves more cleanly from the research gap to the methodology, then to the selected-surrogate comparison, and finally to the

Comment II.18:

The English is not fully consistent and is difficult to follow in several places, starting from the abstract. Some sentences are overly complex and do not read well. A thorough language revision is needed to improve clarity, consistency, and readability throughout the manuscript.

Answer: We agree. The manuscript has undergone a full language revision in the current round. This includes the abstract, introduction, discussion, appendix, and figure captions. The current wording is also more internally consistent because the paper now follows one bounded storyline throughout: benchmark fragility under surrogate conditioning rather than optimizer-superiority rhetoric.

In addition to sentence-level smoothing, we also used this revision round to remove several places where the old wording still reflected earlier result bundles or earlier interpretations of the benchmark. As a result, the present manuscript is not only cleaner linguistically, but also more coherent in the way each section supports the same final claim boundary.

Revised manuscript excerpt

From the revised Abstract:

“On the selected 2000-sample surrogate, NSGA-II achieves stronger objective summaries, the largest Hypervolume, and the lowest IGD, while all three DDPG scenarios remain weaker under post-hoc preference-aligned utility selection. Taken together, the paper supports a methodological interpretation of surrogate-conditioned benchmark fragility in urban energy design, not a claim of general policy-learning superiority.”

Comment II.19:

The abstract is concise and generally clear. However, some claims are too strong relative to the level of validation provided in the paper, e.g., phrases such as "substantially outperformed." The wording should be moderated and tied more directly to the reported results. Since the study relies on a surrogate-based environment rather than direct simulation during optimization, this limitation should also be reflected, at least briefly, in the abstract.

Answer: We agree and revised the abstract accordingly. The current abstract no longer uses superiority wording. Instead, it states the selected-surrogate result directly, identifies the comparison as surrogate-conditioned, and notes the bounded physical probe as supporting evidence rather than as full closure.

We also revised the abstract so that each major sentence more clearly serves one of four functions: background and motivation, benchmark setup, core result, and claim boundary. This was done to ensure that the abstract no longer reads like a compressed positive-results summary, but rather like a balanced summary of what the revised evidence actually supports. In particular, the revised abstract now reports the final method ordering explicitly and places the validation limitation in the abstract itself rather than leaving it entirely to the discussion.

Revised manuscript excerpt

The revised Abstract now states:

“On the selected 2000-sample surrogate, NSGA-II achieves stronger objective summaries, the largest Hypervolume, and the lowest IGD, while all three DDPG scenarios remain weaker under post-hoc preference-aligned utility selection. A small-batch physical probe adds bounded external support: EUIt aligns most consistently, the current EG check remains weaker but directionally stable, and a revised windowsill-style direct-sun-hours check places H in the same broad range as the surrogate values, albeit with weaker stability than EUIt.”

Comment II.20:

The introduction has a clear storyline, but the main methodological argument needs to be better justified. The key issue is not simply whether DQN is unsuitable, but whether DDPG is the most appropriate continuous-control method for this problem compared with alternatives (e.g., TD3, SAC, PPO with continuous actions, CMA-ES, MOEA/D, or Bayesian optimization). What I can see from the current version is that the paper argues mainly for DRL pros vs traditional optimization cons, but this comparison is not framed broadly enough. The discussion of DRL versus NSGA-II should be strengthened. The current framing emphasizes the limitations of NSGA-II and the promise of DRL, but it does not sufficiently discuss the broader set of state-of-the-art baselines for continuous multi-objective optimization. A more balanced positioning would improve the technical credibility of the paper.

Answer: We agree. The revised Introduction now positions DDPG as an intentionally narrow policy-learning baseline rather than as the presumptively best continuous optimizer for this task. The literature review now discusses stronger actor-critic alternatives such as SAC, TD3, PPO, and TRPO, and it also positions CMA-ES as a relevant derivative-free baseline from neighboring energy-system optimization work. In addition, Appendix B now includes a same-budget CMA-ES follow-up, while the manuscript makes clear that SAC is discussed from the literature only and is not executed in the current work. This broader positioning improves the technical credibility of the comparison without overstating the current evidence.

This was an important change in the revised manuscript because the paper is no longer written as if the relevant comparison were only “DRL versus NSGA-II.” Instead, the current text explains why DDPG is used here as a deliberately narrow policy-learning test case, while the appendix demonstrates that even a stronger continuous optimizer such as CMA-ES can exploit the same surrogate corner artifact under the same budget. This allows the manuscript to separate DDPG-specific instability from the broader question of surrogate-conditioned benchmark fragility.

Revised manuscript excerpt**Revised literature-positioning text:**

“The present paper should be interpreted as an intentionally narrow DDPG-versus-NSGA-II study within one surrogate-conditioned urban-design workflow, not as a claim that the compared pair exhausts the relevant optimizer space. In particular, SAC and other entropy-regularized actor-critic variants are discussed here only as relevant external baselines from the literature; they are not executed in the current work and therefore should not be folded into the paper’s causal claims about deterministic policy-gradient behavior.”

Comment II.21:

I suggest revising the section title "Relevant work" to a more standard title such as "Background" or "Literature review." In addition, the statement "this study introduces DRL" should be rephrased as 'adopted' or 'applied'.

Answer: We thank the Reviewer for this suggestion. This has been implemented. The relevant section is now titled “Literature Review and Research Gap,” and the wording has been revised so that DRL is presented as adopted in the current methodological setting rather than introduced as if it were novel in itself.

Although this change is brief in wording, it matters for the paper’s technical posture. The revised phrasing helps avoid overstating novelty at the algorithm level and better aligns the manuscript with its actual contribution, which is methodological comparison and diagnosis within a specific surrogate-conditioned urban-design workflow. This comment helped us remove wording that was stronger than the paper’s real contribution.

Revised manuscript excerpt**Revision summary:**

The corresponding section is now titled “Literature Review and Research Gap,” and the revised wording consistently refers to adopting or using DDPG in the present workflow.

Comment II.22:

I appreciate the authors' effort to articulate the novelty. However, in the current version, the novelty statement is somewhat repetitive and slightly overstated. In my view, the most credible contribution is not DRL alone, but the use of a policy-learning framework for urban morphology optimization under different reward priorities. The novelty points could be merged and stated more compactly.

Answer: We agree. The contribution list has been rewritten in a more compact and restrained way. The revised version now emphasizes four points only: the surrogate-assisted policy-learning formulation, the selected-surrogate comparison against fair-budget NSGA-II, the interpretation of the resulting ranking as a benchmark-fragility problem, and the morphology-level plus validation-mode diagnostics used to bound the claims.

We intentionally compressed the contribution list because the earlier version risked splitting one underlying contribution into multiple overlapping claims. The revised list is designed to make the paper's actual contribution easier to recognize: not "many independent novelties," but one tightly bounded study whose value comes from the combination of benchmark design, diagnostic analysis, and explicit claim control. We believe this revised form is more credible and more in line with the contribution that the current evidence can genuinely support.

Revised manuscript excerpt
Revised contribution list:

The current contribution list now focuses on the DRL-DNN framework, the finding that fair-budget NSGA-II remains stronger on the selected highest-accuracy surrogate, the benchmark-fragility interpretation, and the bounded diagnostic role of morphology analysis plus reevaluation and physical probing.

Comment II.23:

I acknowledge that the procedure used to generate the generic urban morphologies is technically sound, and the supplementary material is helpful. However, the resulting database appears limited relative to the size of the design space. The study uses 12 Urban Morphology Factors (UMFs), which defines a 12-dimensional design space, but the surrogate seems to be trained on only 500 cases, inferred from the 5-fold cross-validation description on page 9: "4 folds (400 samples) were used for training, and the held-out 1-fold (100 samples) was used for testing." If this interpretation is correct, the manuscript should state this explicitly in Section 2.A. At present, the actual number of generated urban morphologies is not presented clearly enough. This is important because a limited dataset in a 12-dimensional space may restrict the surrogate's reliability, especially near optima.

Answer: We agree and clarified this point. The revised manuscript now states the scale-study protocol explicitly: the workflow examines 500, 1000, 1500, and 2000 generated morphologies, and the final optimizer benchmark is conducted on the selected 2000-sample surrogate from that study. This makes the sample-size story and the final comparison basis much clearer than before.

In other words, the current revision no longer leaves the reader with the impression that the final benchmark rests directly on the original 500-sample baseline. The selected benchmark now sits on the best-performing surrogate retained from the scale study, while the manuscript continues to acknowledge that even the selected 2000-sample regime remains limited relative to the full 12-dimensional design space.

Revised manuscript excerpt
Revised methodology text:

"A dataset-scale study was carried out over 500, 1000, 1500, and 2000 morphologies, and the final optimizer comparison uses the most accurate surrogate identified from that study."

Comment II.24:

The DNN framework is described clearly at a general level, but the key concern is not the architecture itself. The main issue is whether the surrogate is sufficiently reliable for optimization. A dataset of about 500 samples is limited for a 12-variable problem, particularly when the DRL agent is expected to explore extreme regions of the design space. The paper would be much stronger if the final best solutions identified by DRL and NSGA-II were re-evaluated using the original simulation workflow, rather than relying only on surrogate predictions.

Answer: We agree. The revised package strengthens this point in three ways. First, Appendix A now reports the scale-study diagnostics, cross-validation summary, and additional surrogate-fidelity and tail-region diagnostics more explicitly. Second, Appendix B now reports independent deterministic reevaluation of the selected candidates from the highest-accuracy 2000-sample comparison. Third, Appendix B now also includes a bounded physical probe on representative candidates. We state explicitly that this still does not replace full publication-grade physical closure, but it does move the paper beyond pure optimization-time surrogate reporting.

We also use the revised discussion to make the reliability boundary explicit rather than implicit. The manuscript now states that surrogate fit quality alone is not sufficient for optimization-facing confidence, because benchmark ordering can still change across checkpoints and candidate-level drift remains visible after independent reevaluation. This was added to avoid giving the impression that the surrogate-validation package by itself closes the optimization-reliability question.

Revised manuscript excerpt

Revised appendix text:

“To reduce over-reliance on surrogate predictions, the workflow re-evaluates preference-aligned candidate solutions using an independent analytic simulation call that follows the same response definitions used to generate the surrogate-training data.”

Revised physical-probe text:

“The purpose of this table is not to claim full physical closure, but to show that the physical probe has moved beyond proof-of-life and into a partially aligned diagnostic regime.”

Comment II.25:

The DRL framework is interesting, but the formulation should be justified more carefully. The problem being solved is essentially a static black-box optimization problem, while the RL formulation implies a sequential decision process. Since the next state is deterministically generated by the surrogate from the selected action, the sequential structure appears weak. The paper should explain more clearly why DDPG is preferable to direct continuous optimization methods in this setting. In addition, the reward formulation is based on a weighted distance to a utopian point, which effectively turns the problem into a scalarized single-objective optimization, rather than direct Pareto learning.

Answer: We agree. The revised manuscript now treats this point directly and more conservatively. We state explicitly that the present task is a scalarized static black-box optimization problem recast as policy learning, and we do not argue that DDPG is preferable to direct continuous optimization methods in general. We also clarify that the reward is a weighted-distance scalarization around a utopian point rather than a direct Pareto-learning objective. This clarification is now part of both the method description and the discussion of why DDPG underperforms NSGA-II on the selected surrogate.

We considered this a particularly important criticism because it affects the conceptual framing of the whole paper rather than only one formula or one paragraph. For that reason, the revised manuscript now addresses this issue in more than one place: first in the method description, where the scalarized nature of the reward is stated explicitly; and then again in the discussion, where the mismatch between static black-box optimization and actor-critic training dynamics is used to interpret the final benchmark result more honestly.

Revised manuscript excerpt

Revised discussion text:

“The present RL formulation is a scalarized static black-box optimization problem recast as policy learning; therefore, alternative continuous optimizers remain important baselines for future work.”

Comment II.26:

The reward depends on min and max values over the feasible domain. The authors should explain how these bounds are defined, and whether they come from the sampled dataset or from known design limits. If they are derived from the sampled dataset, then the reward formulation may depend strongly on sample coverage.

Answer: We thank the Reviewer for raising this point. It is now clarified in the reward subsection. The manuscript states explicitly that the normalization bounds are derived from the sampled design domain represented in the surrogate-training dataset. We also note there that this makes the reward scale dependent on dataset coverage and should therefore be interpreted as a practical normalization choice rather than as a uniquely defined theoretical bound.

We considered this clarification important because it ties the reward formulation back to the paper’s broader limitation structure. Once the normalization depends on the sampled domain, the resulting optimization behavior must also be interpreted relative to that domain coverage, rather than as if the reward were defined over a fully known global design space.

Revised manuscript excerpt

Revised reward text:

“The normalization bounds are derived from the sampled design domain represented in the surrogate-training dataset. This choice makes the reward scale dependent on dataset coverage and should therefore be interpreted as a practical normalization choice rather than as a uniquely defined theoretical bound.”

Comment II.27:

For Eq 6, the statement that the reward was "designed with an upper limit of 10^6 to ensure convergence" sounds ad hoc and requires clarification. A brief practical explanation would be enough.

Answer: We agree and appreciate this request for clarification. The wording has been revised accordingly. The manuscript now states that the factor 10^6 is only a numerical offset used to keep rewards positive and stabilize optimization logs. It does not alter the ranking induced by the weighted-distance objective itself.

We made this change because we agreed that the earlier wording could sound more theoretically motivated than it actually was. The revised explanation is intended to be operationally transparent: the factor is a numerical convenience, not a substantive modeling claim.

Revised manuscript excerpt

Revised explanation:

“The factor 10^6 is used only as a numerical offset to keep rewards positive and to stabilize optimization logs; it does not alter the ranking induced by the weighted distance.”

Comment II.28:

Since one of the main conclusions is that DRL outperforms NSGA-II, the benchmarking protocol needs to be more transparent. Authors should clearly report whether the two methods used the same evaluation budget, stopping criteria, and surrogate environment, and whether the comparison is based on multiple independent runs (and also computational efficiency).

Answer: We agree that the benchmark protocol needed to be more transparent, and we revised it substantially. The current

manuscript now reports the selected-surrogate comparison under a matched per-seed interaction budget, matched seed count, and a shared surrogate artifact. It also reports a representative local CPU-only wall-clock and memory audit. Most importantly, the premise of the comment no longer applies in the revised manuscript: the paper does not conclude that DRL outperforms NSGA-II. The current result is instead that fair-budget NSGA-II remains stronger than all three DDPG scenarios on the selected 2000-sample surrogate.

To make the revised comparison easier to inspect, the manuscript now contains both a dedicated budget-accounting table and a benchmark summary figure. The resulting comparison is therefore transparent at the levels of query budget, seed count, result ordering, and representative compute cost, rather than relying on scattered descriptions across sections. We also found it important to state the remaining asymmetry explicitly, namely that DDPG depends on policy training while NSGA-II does not, so that the compute-efficiency interpretation is not overstated.

Revised manuscript excerpt

Revised benchmark-accounting text:

“The comparison matches both the per-seed interaction budget and the number of seeds between DDPG and NSGA-II, while also making clear that DDPG depends on policy training whereas NSGA-II does not.”

Revised result text:

“On the selected 2000-sample surrogate, fair-budget NSGA-II remains stronger than all three DDPG scenarios in both archive-level Pareto metrics and preference-aligned utility.”

../elsarticle/fig/fig9.pdf

Revised benchmark summary figure now used in the manuscript to make the final comparison protocol and method ordering visible at a glance.

Comment II.29:

The paper clearly acknowledges major limitations. However, some of the above methodological limitations, especially those related to surrogate reliability and benchmarking fairness, should also be reflected more clearly in the discussion.

Answer: We agree. The Discussion section has been rewritten substantially. It now states explicitly that the comparison should be interpreted conditionally because it remains surrogate-conditioned, checkpoint-sensitive, and only partially validated outside the analytic workflow. It also states clearly that NSGA-II remains stronger than the tested DDPG setup on the selected surrogate, that DDPG remains unstable, that the bounded physical probe does not replace the main surrogate benchmark, and that broader baselines such as SAC remain future work.

This revised discussion is intentionally more explicit than the previous version. Instead of relegating the main caveats to scattered side remarks, the current text gathers them in one place: surrogate-conditioning, candidate drift after reevaluation, checkpoint-dependent ranking changes, incomplete physical closure, limited compute auditing, and the fact that broader actor-critic baselines remain future work. Our goal in doing so was to ensure that the final paper reads as a careful and technically bounded study rather than as a result that is stronger than the current evidence allows.

Revised manuscript excerpt

Revised limitations text:

“The current comparison remains surrogate-based during optimization, benchmark ordering itself can change across checkpoint selections, a representative local CPU-only wall-clock and memory audit is now available but is still a single-machine reference, and although the current comparison now includes a CMA-ES follow-up, entropy-regularized actor-critic baselines such as SAC remain untested here.”